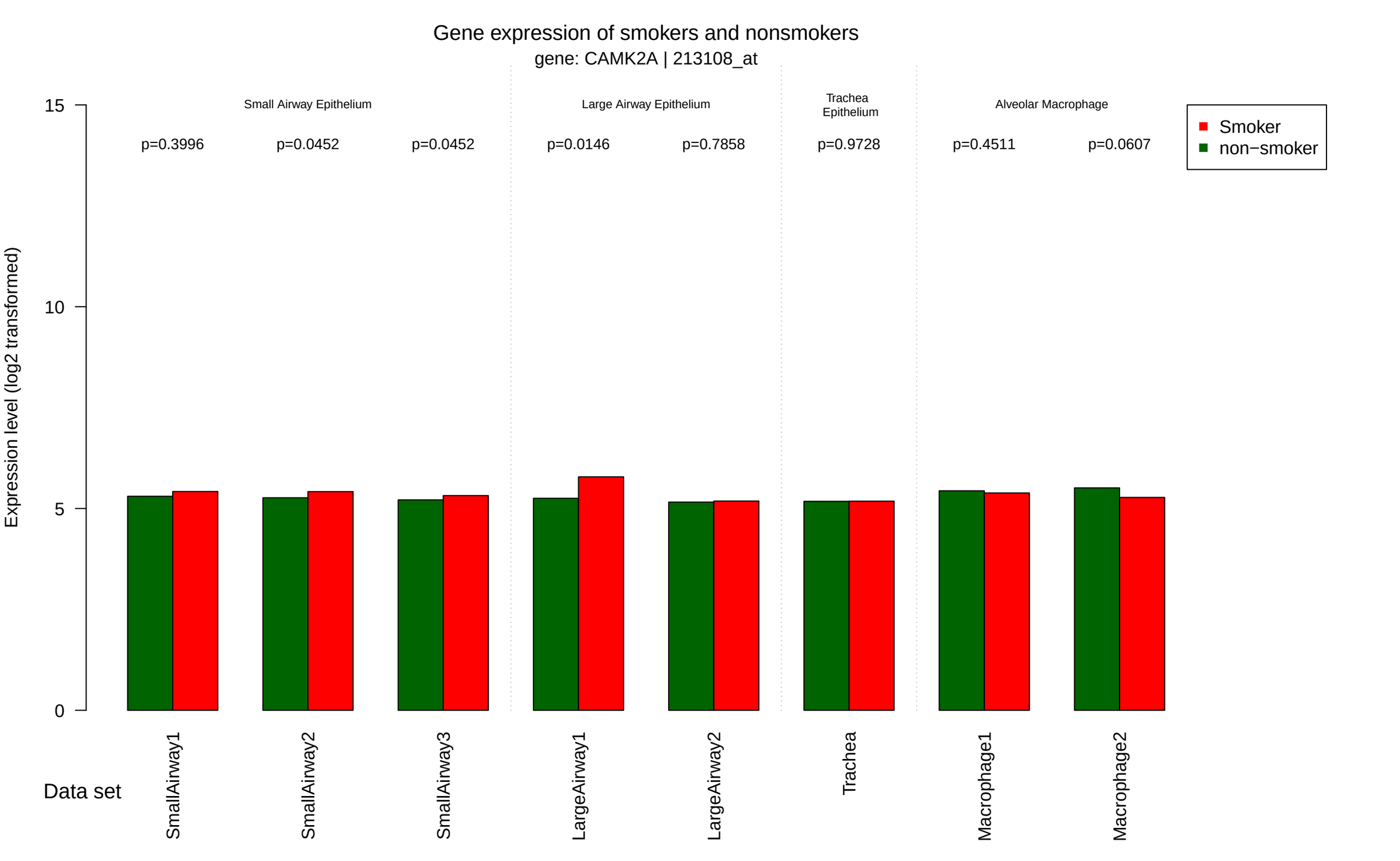
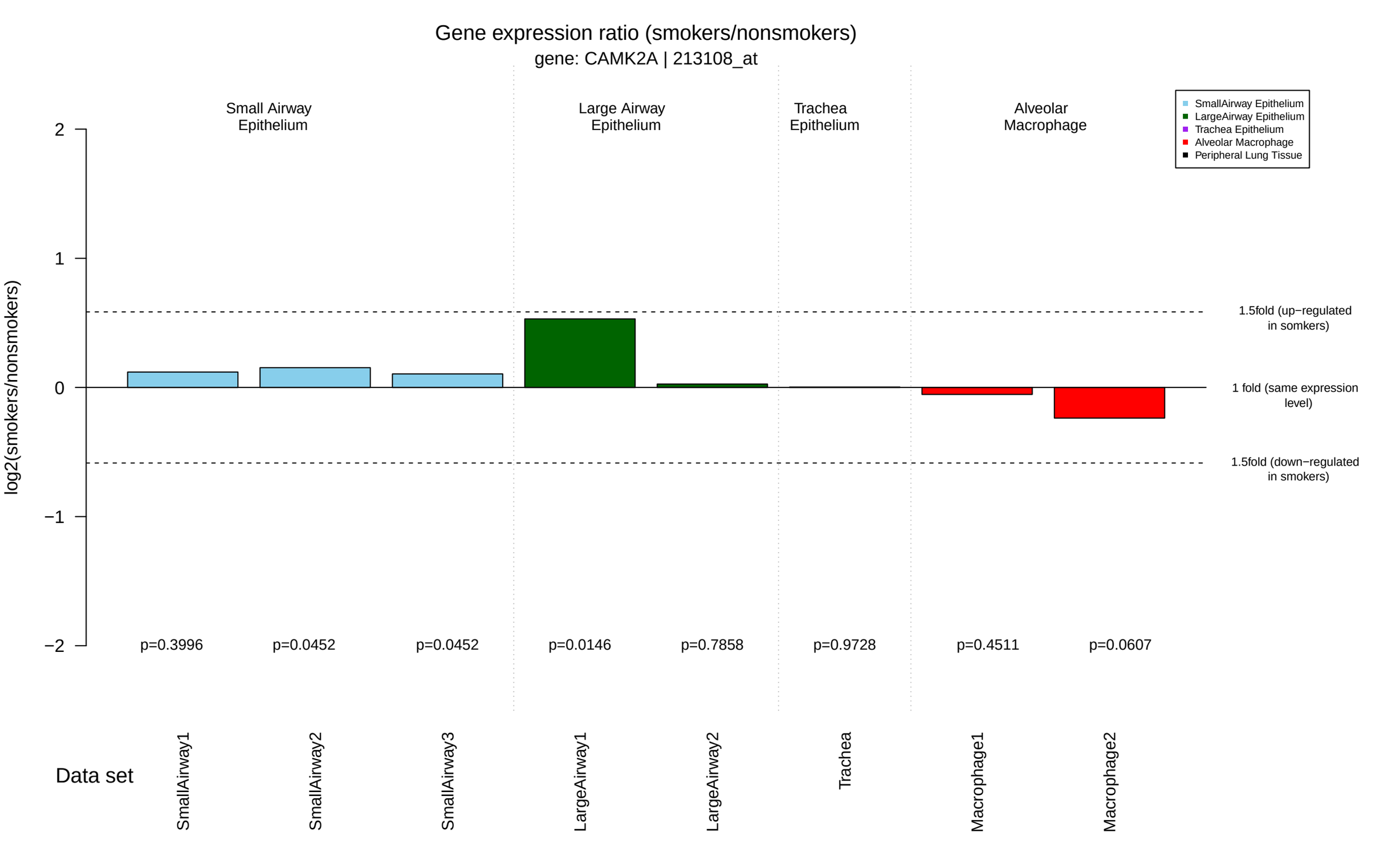
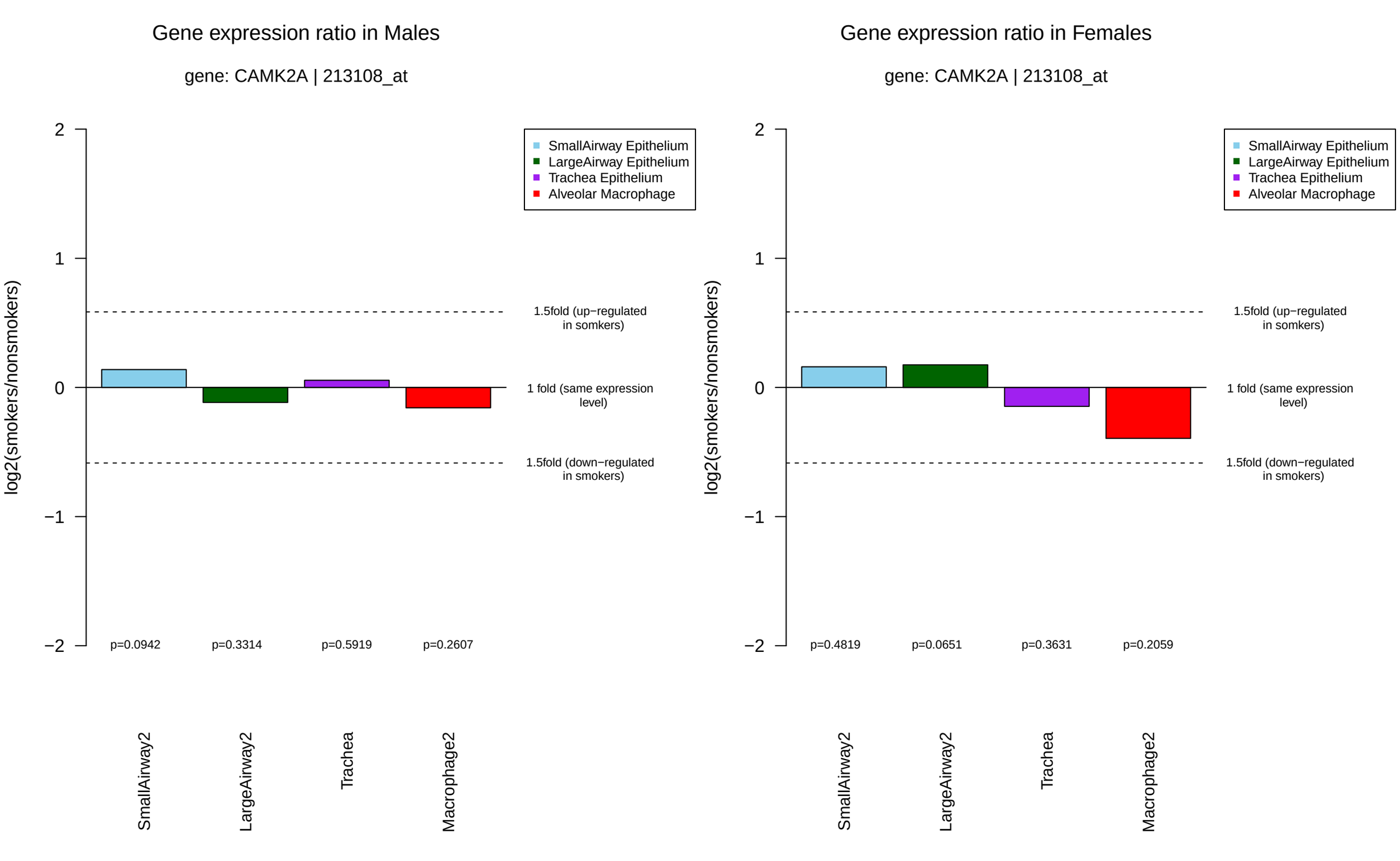
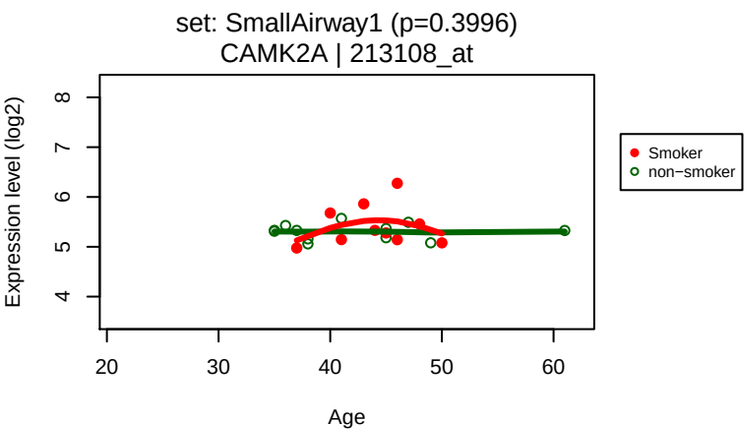
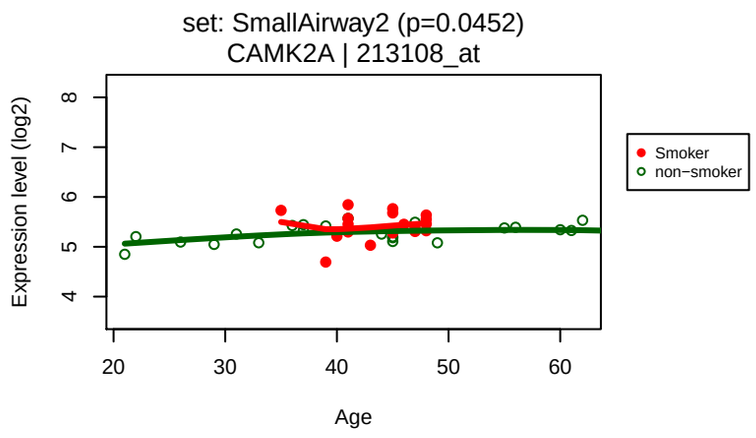
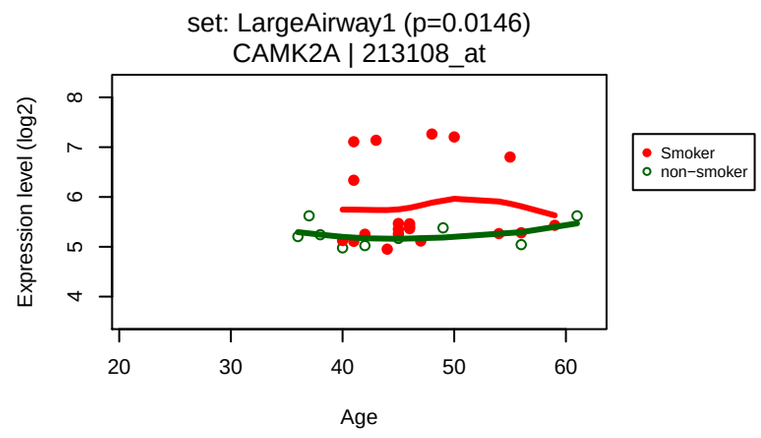

Gene expression of smokers and nonsmokers
gene: CAMK2A | 213108_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3396
p=0.0452
p=0.0452
p=0.0146
p=0.7858
p=0.9728
p=0.4511
p=0.0607
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

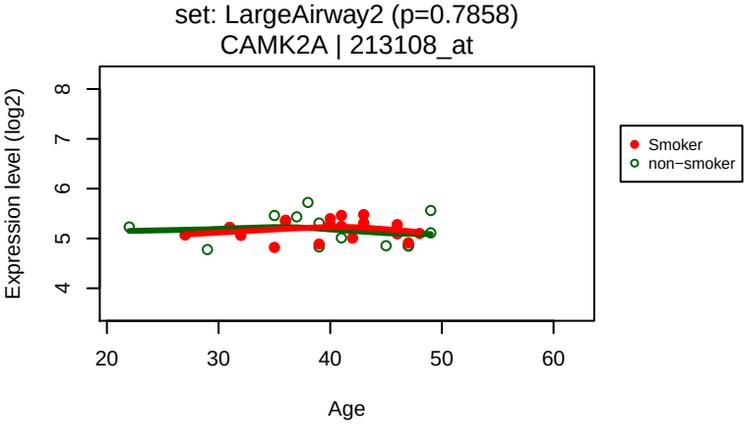

Gene expression ratio (smokers/nonsmokers)
gene: CAMK2A | 213108_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3996
p=0.0452
p=0.0452
p=0.0146
p=0.7858
p=0.9728
p=0.4511
p=0.0607
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

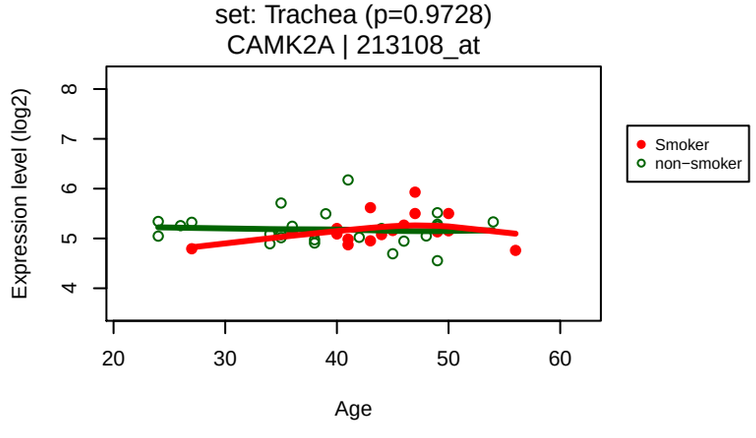

Gene expression ratio in Males
gene: CAMK2A | 213108_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0942
p=0.3314
p=0.5919
p=0.2607
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CAMK2A | 213108_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4819
p=0.0651
p=0.3631
p=0.2059
SmallAirway2
LargeAirway2
Trachea
Macrophage2

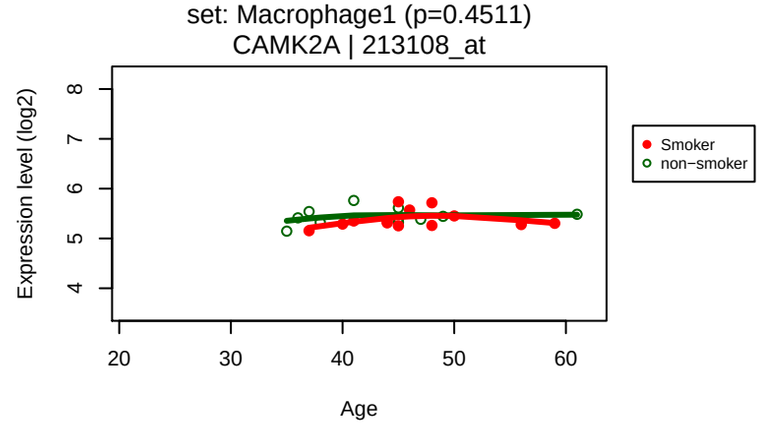

set: SmallAirway1 (p=0.3996)
CAMK2A | 213108_at
Expression level (log2)
Age
Smoker
non−smoker

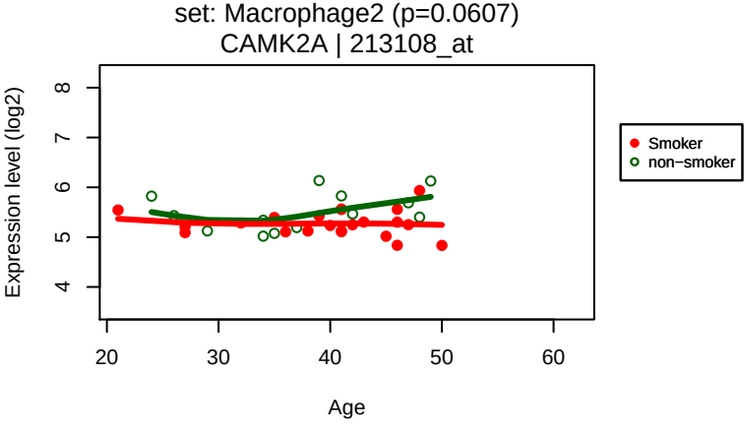

set: SmallAirway2 (p=0.0452)
CAMK2A | 213108_at
Expression level (log2)
Age
Smoker
non−smoker

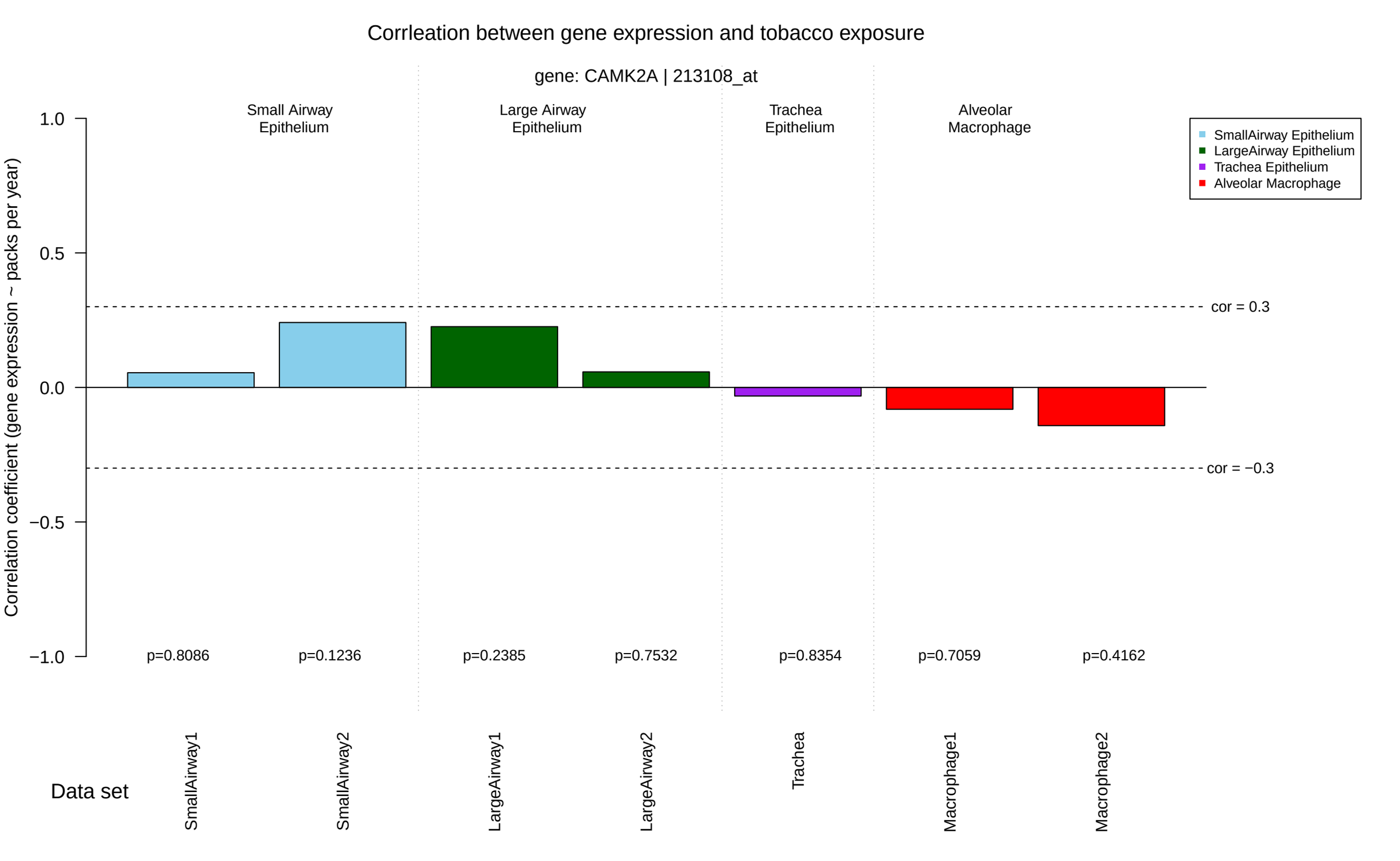

set: LargeAirway1 (p=0.0146)
CAMK2A | 213108_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7858)
CAMK2A | 213108_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9728)
CAMK2A | 213108_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4511)
CAMK2A | 213108_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0607)
CAMK2A | 213108_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CAMK2A | 213108_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8086
p=0.1236
p=0.2385
p=0.7532
p=0.8354
p=0.7059
p=0.4162
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2